

Cell Design Specification - VBF-T8R2-DS-01 (t8_r2, final design V_G)

1. Basic specification

Item	Value	Source
Electrochemical system	NMC811 / graphite (Chen2020)	parameter set
Nominal capacity (Ah)	5.000	parameter set
Simulated 1C capacity (Ah)	5.0351	cell/r3_VG_1c_dfn.json
Voltage window (V)	2.5 - 4.2	parameter set
Dimensions (mm)	height 65.0 x width 1990.8 x stack 0.1506; shell: Not provided	parameter set + cell/r3_VG_energy.json
Electrolyte transport	D 4.20e-10 m2/s; t+ 0.400; sigma 1.20 S/m; Li0 1000 mol/m3	parameter set (Chen2020 base + cell/params_r3_VG.json overrides)

2. Electrode and separator

Layer	Thickness (um)	Porosity	Collector
Positive (NMC811)	60.0	0.335	Al 8.0 um
Negative (graphite)	67.62	0.250	Cu 6.0 um
Separator	9.0	0.470	-
Particle radii (um)	pos 3.0 / neg 3.5	-	V_G microstructure lever

$$N/P \text{ (mechanical)} = (c_max_neg \times amf_neg \times th_neg) / (c_max_pos \times amf_pos \times th_pos) = 0.6674$$

3. Process design parameters

Parameter	Value	Formula
Positive areal density (g/m2)	130.15	thickness x (1-porosity) x density x 1000
Negative areal density (g/m2)	84.03	thickness x (1-porosity) x density x 1000
Positive compaction density (g/cm3)	2.1692	electrode density x (1-porosity) / 1000
Negative compaction density (g/cm3)	1.2428	electrode density x (1-porosity) / 1000
Electrolyte fill (g)	6.403	pore volume x 1200 kg/m3 (literature value)

4. Mass breakdown

Layer	Mass (g)
Positive coating	16.842
Negative coating	10.874
Al collector	2.795
Cu collector	6.957
Separator	0.245
TOTAL (contract caliber)	37.713
Electrolyte (annotated, excluded)	6.403

5. Performance verification (vs log entry 0 criteria)

Item	Result	Criterion	Determination
Energy density (Wh/kg)	483.4	>= 446.18	PASS
5C retention	0.9631	>= 0.90	PASS
Cell mass (g)	37.71	<= 40	PASS
4C-charge plating	anode min 0.0215 V > 0	plated == false	PASS
4C-charge temperature (K)	351.93 (+33.78 K)	none (record)	record

6. Design notes

Baseline: ED 400.75 Wh/kg, retention 0.0874, mass 43.45 g, plating at 4C. R2 transport lift: retention 0.676-0.722 (salt depletion resolved). R2 diagnosis: positive solid-phase surface depletion ($\tau_{\text{diff}} \sim 6812$ s). R3 (particle radius 3.0/3.5 μm + $t_{\text{+}}/D$ margin = V_{G}): all four criteria PASS. See log.jsonl rounds 1-3 and final entry.